Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

WSQ Technical Drawing with AutoCAD

TGS Ref No: TGS-2021005538

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 13

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
11	18 August 2026	Legacy release — master trainer slides v11 with 2-day schedule.	Dr Alfred Ang
12	18 August 2026	Full single-source regeneration: 151-slide house-style deck, 16 hands-on labs, 2-day schedule and aligned slide references.	Dr Alfred Ang
13	21 August 2026	DWG integration release: Labs 1, 8–12, 15 and 16 now use verified practice drawings; the Lab 12 relative Xref dependency is complete; detailed procedures remain in the guide/labs while the deck uses visual briefings and verification slides. Slide references regenerated for v13.	Dr Alfred Ang

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Course Information

Course Title	WSQ Technical Drawing with AutoCAD
WSQ Course Reference	TGS-2021005538
Skills Framework TSC	Technical Drawing (DSN-TDR-3005-1.11)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours, including assessment)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on AutoCAD drafting labs — one computer with AutoCAD per learner
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Read and visualize AutoCAD technical drawings
- LO2: Produce 2D and 3D technical drawings using AutoCAD
- LO3: Create part lists using AutoCAD
- LO4: Create AutoCAD design documentation
- LO5: Produce design specification using AutoCAD

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 1.5 hours, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:00 pm (WA 60 min, PP 90 min).
- A minimum of 75% attendance (SSG Digital Attendance record) and a 'Competent' assessment result are required for funding.

Course Schedule

Day 1 — AutoCAD Foundations, Drawing & Modifying Objects

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, ground rules, trainer and learner introductions, learning outcomes, course outline and mandatory digital attendance (AM)	Slides 3–19

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Time	Duration	Topic / Activity	Slides
10:00–10:45	45 min	**Topic 1 — Getting Started with AutoCAD: user interface, workspaces, model/paper space, navigation, precision aids, units (concepts + demo)**	Slides 20–31
10:45–11:15	30 min	Hands-on: Lab 1: Setting Up a New Drawing	Slides 32–34
11:15–11:30	15 min	Tea break	
11:30–12:30	60 min	**Topic 2 — Technical Drawing with AutoCAD: lines, circles, rectangles, polygons, ellipses, arcs, polylines, splines, construction geometry, isometric drawing (concepts + demo)**	Slides 35–43
12:30–13:30	60 min	Lunch break	
13:30–15:30	120 min	Hands-on: Lab 2: Creating Lines and Coordinates; Lab 3: Circles, Rectangles and Polygons; Lab 4: Polylines, Arcs, Ellipses and Splines; Lab 5: 2D Isometric Drawing	Slides 44–52
15:30–15:45	15 min	Tea break	
15:45–16:30	45 min	**Topic 3 — Modifying Objects: move/copy/rotate/offset, mirror/stretch/scale, trim/extend, fillet/chamfer, arrays (concepts + demo)**	Slides 53–62
16:30–18:15	105 min	Hands-on: Lab 6: Move, Copy, Rotate and Offset; Lab 7: Trim, Extend, Fillet and Chamfer; Lab 8: Mirror, Stretch, Scale and Arrays	Slides 63–69
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance	

Total training time: 480 minutes (8 hours).

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Day 2 — Organizing Objects, Design Documentation & Assessment

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)	
9:45–10:25	40 min	**Topic 4 — Organizing Objects: object properties, layers, grips, hide/isolate, blocks & attributes, external references (concepts + demo)**	Slides 70–78
10:25–11:10	45 min	Hands-on: Lab 9: Object Properties and Layers; Lab 10: Grips, Hide and Isolate	Slides 79–82
11:10–11:25	15 min	Tea break	
11:25–12:15	50 min	Hands-on: Lab 11: Blocks, Attributes and Part Lists; Lab 12: External References and PDF Import	Slides 83–87
12:15–13:15	60 min	Lunch break	
13:15–13:55	40 min	**Topic 5 — Design Documentation: text & styles, dimensions, multileaders, tables & part lists, layouts, viewports, plotting, hatching (concepts + demo)**	Slides 88–98
13:55–14:40	45 min	Hands-on: Lab 13: Text and Text Styles; Lab 14: Dimensions, Multileaders and Part-List Tables	Slides 99–102
14:40–14:55	15 min	Tea break	
14:55–15:30	35 min	Hands-on: Lab 15: Hatching and Detail Finishing; Lab 16: Layouts, Viewports and Plotting	Slides 103–107
15:30–15:45	15 min	Course recap, feedback, TRAQOM survey and assessment-readiness briefing	Slides 108–112
15:45–16:00	15 min	**Briefing for Assessment and Assessment digital attendance**	
16:00–17:00	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 60 minutes, open book**	

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Time	Duration	Topic / Activity	Slides
17:00–18:30	90 min	**Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 90 minutes, open book. PM digital attendance**	

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the Skills Framework TSC)

Topic / TSC mapping	Ability · Knowledge	Labs
Topic 01: Getting Started with AutoCAD	A1 · K1	Lab 1
Topic 02: Technical Drawing with AutoCAD	A2 · K2	Lab 2, Lab 3, Lab 4, Lab 5
Topic 03: Modifying Objects	A3 · K3	Lab 6, Lab 7, Lab 8
Topic 04: Organizing Objects	A4 · K4	Lab 9, Lab 10, Lab 11, Lab 12
Topic 05: Design Documentation	A5 · K5	Lab 13, Lab 14, Lab 15, Lab 16

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